

A Finite Mathematical Framework Correlating Cognitive Behavioral Optimization, Pharmacological Mediators, and QML Neuromodulation in Schizoaffective Disorder

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April 2026

Abstract

We introduce a finite deterministic mathematical framework to model the complex neural correlates of Schizoaffective Disorder under the influence of both pharmacological antipsychotics and limited intake of <4.5% ABV beer. Traditional approaches treat alcohol unconditionally as a rigid trigger; here, we quantify the precise boundaries where mild social reward circuits can be activated without destabilizing prefrontal gating via mesolimbic dopamine surge. Furthermore, we outline the integration of Quantum Machine Learning (QML) optimized Transcranial Magnetic Stimulation (TMS), alongside Hebbian amplification metrics and Cortical Control indices, explicitly formulated to predict and minimize relapse probability while maximizing Cognitive Behavioral Therapy (CBT) outcomes.

1. Dopamine Variance & Neuropharmacological Bounding

The primary driver of psychotic manifestation in the model is the variance in dopamine, σ_d^2 , influenced linearly by beer intake units B and inversely by medication units M .

$$\sigma_d^2 = \max(0.1, \sigma_{base}^2 + 1.5(0.35B) - 1.2M)$$

This variance dictates the probability density function (PDF) of symptom severity x :

$$P(x) = \frac{1}{\sigma_d \sqrt{2\pi}} e^{-\frac{(x - \mu_{symp})^2}{2\sigma_d^2}}$$

2. Prefrontal Cortical Control Index (C_{pfc})

To quantify executive oversight, we take the ratio of prefrontal node activity (A_{pfc}) to subcortical limbic hyperactivity (average of Amygdala A_{amy} & Hippocampus A_{hip}):

$$A_{pfc} = \max(0.1, 1.0 - 0.8(0.35B) + 0.5M)$$

$$C_{pfc} = \min\left(1.0, \frac{A_{pfc}}{\max(0.1, \frac{1}{2}(A_{amy} + A_{hip}))}\right)$$

3. Hebbian Amplification Optimization for CBT

Neuroplastic efficiency underlying cognitive restructuring is maximized when cognitive restructuring dominates and dopamine variance is stable. The Hebbian Amplification Efficiency H is defined as:

$$H(\%) = \min\left(100.0, \max\left(0.0, \left[\frac{A_{pfc}}{\sigma_d^2}\right] \times 45.0\right)\right)$$

4. Minimal Relapse Probability Derivation

The deterministic probability of clinical relapse (P_{rel}) emerges as a composite vector of model variables:

$$P_{rel} = 100.0 - 80.0(C_{pfc}) + 15.0(\sigma_d^2) - 10.0(M) - \Delta_{CBT}$$

Where $\Delta_{CBT} = 15.0$ iff the composite CBT Optimization Score ≥ 70 .

5. QML Neuromodulation Trajectories (TMS)

Quantum state measurements of entangled subsystem proxies are iteratively optimized to derive the exact therapeutic frequency for transcranial magnetic stimulation

$$F_{TMS} = 10.0 + 8.5 \cdot \sigma_d^2 \text{ (Hz)}$$

This frequency dynamically tracks neurotransmitter volatility to enforce phase-lock entrainment across the frontostriatal tracts. Combined with precise pharmacological dosing, the overall projected yield equation resolves as:

$$\gamma_{yield}(\%) = \min(99.9, \max(0, 65.0 - 30.0(0.35B) + 45.0M))$$

